

Supplementary Derivation for the Paper Fault-Aware Design for Reconfigurable Holographic Surface-Aided ISAC Systems

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This supplementary document provides the detailed derivations of $\mathbf{A}_{\eta_0}$ and $\mathbf{B}_{\eta_0}$ used for the paper L. Wang, M. Delbari, G. Zhou, L. F. Abanto-Leon, M. Hollick, and V. Jamali, “Fault-Aware Design for Reconfigurable Holographic Surface-Aided ISAC Systems,” *2026 IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2026. These derivations are omitted from the main paper due to space limitations.

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Note that the equation numbering in this document follows the numbering sequence used in the main paper. Detailed derivations are shown below.

The calculation of MCRB is dependent on the derivation of $\mathbf{A}_{\eta}$ and $\mathbf{B}_{\eta}$ given in (14) and (15) in the main paper, in which the first-order and second-order partial derivatives of $\tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})$ with respect to $\boldsymbol{\eta}$ are needed. To achieve this goal, $\tilde{\boldsymbol{\mu}}(\boldsymbol{\eta}) = \text{vec}(\mathbf{G}\tilde{\boldsymbol{\Psi}}\mathbf{Q}\mathbf{X})$ is further written as follows.

$$\tilde{\boldsymbol{\mu}}(\boldsymbol{\eta}) = \alpha \text{vec} \left(\mathbf{a}(\phi_e, \phi_a) \mathbf{a}^H(\phi_e, \phi_a) \tilde{\boldsymbol{\Psi}} \mathbf{Q} \mathbf{X} \right). \quad (32)$$

Denoting $\mathbf{\Omega} = \mathbf{a}(\phi_e, \phi_a) \mathbf{a}^H(\phi_e, \phi_a)$ and $\boldsymbol{\eta} = [\boldsymbol{\phi}^T, \tilde{\boldsymbol{\alpha}}^T]^T$ where $\boldsymbol{\phi} = [\phi_e, \phi_a]^T$ and $\tilde{\boldsymbol{\alpha}} = [\text{Re}(\alpha), \text{Im}(\alpha)]^T$, the first-order partial derivative is obtained as follows

$$\frac{\partial \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\phi}} = \begin{bmatrix} \alpha \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \\ \alpha \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \end{bmatrix}, \quad (33)$$

$$\frac{\partial \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \tilde{\boldsymbol{\alpha}}} = \text{vec} \left(\mathbf{\Omega} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \begin{bmatrix} 1 \\ j \end{bmatrix}, \quad (34)$$

where $\dot{\mathbf{\Omega}}_{\phi_e}$ and $\dot{\mathbf{\Omega}}_{\phi_a}$ are given as follows

$$\dot{\mathbf{\Omega}}_{\phi_e} = \frac{\partial \mathbf{\Omega}}{\partial \phi_e} = j\kappa \left[d_y \sin(\phi_e) \sin(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Y) \right) - d_z \cos(\phi_e) \left(\text{diag}(\mathbf{r}_Z) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Z) \right) \right], \quad (35)$$

$$\dot{\mathbf{\Omega}}_{\phi_a} = \frac{\partial \mathbf{\Omega}}{\partial \phi_a} = -j\kappa d_y \cos(\phi_e) \cos(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Y) \right), \quad (36)$$

$\mathbf{a}$ stands for $\mathbf{a}(\phi_e, \phi_a)$ for the sake of notational convenience. $\mathbf{r}_Y = [0, 1, \dots, R_y - 1]^T$ and $\mathbf{r}_Z = [0, 1, \dots, R_z - 1]^T$ collect the index of y-axis and z-axis, respectively. Next, we take the second-order partial derivative of $\tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})$ with respect to $\boldsymbol{\eta}$, which can be further partitioned as

$$\frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}^2} = \begin{bmatrix} \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\phi}^2} & \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \tilde{\boldsymbol{\alpha}} \partial \boldsymbol{\phi}} \\ \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\phi} \partial \tilde{\boldsymbol{\alpha}}} & \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \tilde{\boldsymbol{\alpha}}^2} \end{bmatrix}, \quad (37)$$

with three elements in (37) given as follows

$$\frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\phi} \partial \tilde{\boldsymbol{\alpha}}} = \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \tilde{\boldsymbol{\alpha}} \partial \boldsymbol{\phi}} = \begin{bmatrix} 1 \\ j \end{bmatrix} \begin{bmatrix} \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \\ \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \end{bmatrix}^T, \quad (38)$$

$$\frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \tilde{\boldsymbol{\alpha}}^2} = \mathbf{0}_2, \quad (39)$$

$$\frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\phi}^2} = \begin{bmatrix} \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \phi_e^2} & \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \phi_a \partial \phi_e} \\ \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \phi_e \partial \phi_a} & \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \phi_a^2} \end{bmatrix} = \begin{bmatrix} \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_e \phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_a \phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \\ \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_e \phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_a \phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \end{bmatrix}, \quad (40)$$

where $\ddot{\mathbf{\Omega}}_{\phi_e \phi_e}$, $\ddot{\mathbf{\Omega}}_{\phi_e \phi_a}$, $\ddot{\mathbf{\Omega}}_{\phi_a \phi_e}$, and $\ddot{\mathbf{\Omega}}_{\phi_a \phi_a}$ are derived as follows

$$\begin{aligned} \ddot{\mathbf{\Omega}}_{\phi_e \phi_e} &= j\kappa \left[d_y \cos(\phi_e) \sin(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Y) \right) + d_z \sin(\phi_e) \left(\text{diag}(\mathbf{r}_Z) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Z) \right) \right. \\ &\quad \left. + d_y \sin(\phi_e) \sin(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \dot{\mathbf{\Omega}}_{\phi_e} - \dot{\mathbf{\Omega}}_{\phi_e} \text{diag}(\mathbf{r}_Y) \right) - d_z \cos(\phi_e) \left(\text{diag}(\mathbf{r}_Z) \dot{\mathbf{\Omega}}_{\phi_e} - \dot{\mathbf{\Omega}}_{\phi_e} \text{diag}(\mathbf{r}_Z) \right) \right], \end{aligned} \quad (41)$$

$$\begin{aligned} \ddot{\mathbf{\Omega}}_{\phi_e \phi_a} &= \ddot{\mathbf{\Omega}}_{\phi_a \phi_e} = j\kappa \left[d_y \sin(\phi_e) \cos(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Y) \right) \right. \\ &\quad \left. + d_y \sin(\phi_e) \sin(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \dot{\mathbf{\Omega}}_{\phi_a} - \dot{\mathbf{\Omega}}_{\phi_a} \text{diag}(\mathbf{r}_Y) \right) - d_z \cos(\phi_e) \left(\text{diag}(\mathbf{r}_Z) \dot{\mathbf{\Omega}}_{\phi_a} - \dot{\mathbf{\Omega}}_{\phi_a} \text{diag}(\mathbf{r}_Z) \right) \right], \end{aligned} \quad (42)$$

$$\ddot{\mathbf{\Omega}}_{\phi_a \phi_a} = j\kappa \left[d_y \cos(\phi_e) \sin(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Y) \right) - d_y \cos(\phi_e) \cos(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \dot{\mathbf{\Omega}}_{\phi_a} - \dot{\mathbf{\Omega}}_{\phi_a} \text{diag}(\mathbf{r}_Y) \right) \right]. \quad (43)$$

Then, we partition $\mathbf{A}_\eta$ and $\mathbf{B}_\eta$ into 4 blocks, namely

$$\mathbf{A}_\eta = \begin{bmatrix} \mathbf{A}_{\phi\phi} & \mathbf{A}_{\alpha\phi} \\ \mathbf{A}_{\phi\alpha} & \mathbf{A}_{\alpha\alpha} \end{bmatrix}, \mathbf{B}_\eta = \begin{bmatrix} \mathbf{B}_{\phi\phi} & \mathbf{B}_{\alpha\phi} \\ \mathbf{B}_{\phi\alpha} & \mathbf{B}_{\alpha\alpha} \end{bmatrix}. \quad (44)$$

According to (14) and (15) in the main paper, each sub-block matrix in $\mathbf{B}_\eta$ is obtained as (45), (46), (47), and (48). Each sub-block matrix in $\mathbf{A}_\eta$ is obtained as (49), (50), (51), and (52).

$$\mathbf{B}_{\phi\phi} = \frac{2|\alpha|^2}{\sigma_s^2} \text{Re} \begin{bmatrix} \text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_e} \mathbf{F} \dot{\mathbf{\Omega}}_{\phi_e}^H \right) & \text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_e} \mathbf{F} \dot{\mathbf{\Omega}}_{\phi_a}^H \right) \\ \text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_a} \mathbf{F} \dot{\mathbf{\Omega}}_{\phi_e}^H \right) & \text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_a} \mathbf{F} \dot{\mathbf{\Omega}}_{\phi_a}^H \right) \end{bmatrix}, \quad (45)$$

$$\mathbf{B}_{\phi\alpha} = \frac{2}{\sigma_s^2} \text{Re} \begin{bmatrix} \alpha^H \text{tr} \left(\mathbf{\Omega} \mathbf{F} \dot{\mathbf{\Omega}}_{\phi_e}^H \right) & \alpha^H \text{tr} \left(\mathbf{\Omega} \mathbf{F} \dot{\mathbf{\Omega}}_{\phi_a}^H \right) \\ j\alpha^H \text{tr} \left(\mathbf{\Omega} \mathbf{F} \dot{\mathbf{\Omega}}_{\phi_e}^H \right) & j\alpha^H \text{tr} \left(\mathbf{\Omega} \mathbf{F} \dot{\mathbf{\Omega}}_{\phi_a}^H \right) \end{bmatrix}, \quad (46)$$

$$\mathbf{B}_{\alpha\phi} = \mathbf{B}_{\phi\alpha}^T, \quad (47)$$

$$\mathbf{B}_{\alpha\alpha} = \frac{2}{\sigma_s^2} \mathbf{I}_2 \text{tr} \left(\bar{\mathbf{\Omega}} \mathbf{R}_x \bar{\mathbf{\Omega}}^H \right), \quad (48)$$

$$\begin{aligned} \mathbf{A}_{\phi\phi} &= \frac{2}{\sigma_s^2} \text{Re} \begin{bmatrix} \varepsilon(\boldsymbol{\eta})^H \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_e\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \varepsilon(\boldsymbol{\eta})^H \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_a\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \\ \varepsilon(\boldsymbol{\eta})^H \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_e\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \varepsilon(\boldsymbol{\eta})^H \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_a\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \end{bmatrix} - \mathbf{B}_{\phi\phi} \\ &= \frac{2|\alpha|^2}{\sigma_s^2} \text{Re} \begin{bmatrix} \text{tr} \left(\ddot{\mathbf{\Omega}}_{\phi_e\phi_e} \bar{\mathbf{F}} \mathbf{\Omega}^H \right) - \text{tr} \left(\ddot{\mathbf{\Omega}}_{\phi_e\phi_e} \mathbf{F} \mathbf{\Omega}^H \right) & \text{tr} \left(\ddot{\mathbf{\Omega}}_{\phi_a\phi_e} \bar{\mathbf{F}} \mathbf{\Omega}^H \right) - \text{tr} \left(\ddot{\mathbf{\Omega}}_{\phi_a\phi_e} \mathbf{F} \mathbf{\Omega}^H \right) \\ \text{tr} \left(\ddot{\mathbf{\Omega}}_{\phi_e\phi_a} \bar{\mathbf{F}} \mathbf{\Omega}^H \right) - \text{tr} \left(\ddot{\mathbf{\Omega}}_{\phi_e\phi_a} \mathbf{F} \mathbf{\Omega}^H \right) & \text{tr} \left(\ddot{\mathbf{\Omega}}_{\phi_a\phi_a} \bar{\mathbf{F}} \mathbf{\Omega}^H \right) - \text{tr} \left(\ddot{\mathbf{\Omega}}_{\phi_a\phi_a} \mathbf{F} \mathbf{\Omega}^H \right) \end{bmatrix} - \mathbf{B}_{\phi\phi}. \end{aligned} \quad (49)$$

$$\begin{aligned} \mathbf{A}_{\phi\alpha} &= \frac{2}{\sigma_s^2} \text{Re} \begin{bmatrix} \varepsilon(\boldsymbol{\eta})^H \alpha \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \varepsilon(\boldsymbol{\eta})^H \alpha \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \\ \varepsilon(\boldsymbol{\eta})^H j \alpha \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \varepsilon(\boldsymbol{\eta})^H j \alpha \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \end{bmatrix} - \mathbf{B}_{\phi\alpha} \\ &= \frac{2|\alpha|^2}{\sigma_s^2} \text{Re} \begin{bmatrix} \text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_e} \bar{\mathbf{F}} \mathbf{\Omega}^H \right) - \text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_e} \mathbf{F} \mathbf{\Omega}^H \right) & \text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_a} \bar{\mathbf{F}} \mathbf{\Omega}^H \right) - \text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_a} \mathbf{F} \mathbf{\Omega}^H \right) \\ j \left(\text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_e} \bar{\mathbf{F}} \mathbf{\Omega}^H \right) - \text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_e} \mathbf{F} \mathbf{\Omega}^H \right) \right) & j \left(\text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_a} \bar{\mathbf{F}} \mathbf{\Omega}^H \right) - \text{tr} \left(\dot{\mathbf{\Omega}}_{\phi_a} \mathbf{F} \mathbf{\Omega}^H \right) \right) \end{bmatrix} - \mathbf{B}_{\phi\alpha}. \end{aligned} \quad (50)$$

$$\mathbf{A}_{\alpha\phi} = \mathbf{A}_{\phi\alpha}^T, \quad (51)$$

$$\mathbf{A}_{\alpha\alpha} = -\mathbf{B}_{\alpha\alpha}, \quad (52)$$

where

$$\mathbf{F} = \tilde{\Psi} \mathbf{Q} (\mathbf{W} \mathbf{W}^H + \mathbf{R}_s) \mathbf{Q}^H \tilde{\Psi}^H. \quad (53)$$

$$\bar{\mathbf{F}} = \tilde{\Psi} \mathbf{Q} (\mathbf{W} \mathbf{W}^H + \mathbf{R}_s) \mathbf{Q}^H \tilde{\Psi}^H. \quad (54)$$

With all the above derivations, we can obtain both $\mathbf{A}_{\eta_0}$ and $\mathbf{B}_{\eta_0}$ by letting $\boldsymbol{\eta} = \boldsymbol{\eta}_0$.